

Compiler Construction

~ Liveness Analysis ~

Goals

How to precisely catch the liveness of each variable?

If a variable is in register \$1, then we can reuse this register as soon as the variable is *no longer used*.

How can we detect: *no longer used*?

Scopes vs. Liveness

Scopes

- Front-end analysis
- Detect names visibility according to textual rules

Liveness

- Back-end analysis
- Focus on all temporaries (not only user-declared variables)
- Exact computation of which temporaries are used at the same time

Liveness Definition

Definition

A variable is **live** at some point in the code if it holds a value that may be used at a later point.

Example

```
1      a := 0
2  L1:  b := a + 1
3      c := c + b
4      a := b * 2
5      if a < N goto L1
6      return c
```

Question:

What are the liveness of a , b and c ?

Flow Graph

Liveness computation requires an adequate data-structure.

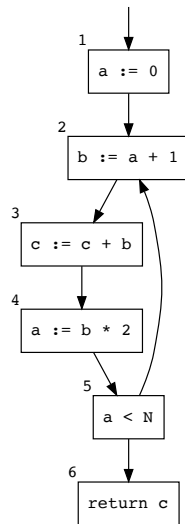
Control Flow graph (CFG):

A representation, using graph notation, of all paths that might be traversed through a program during its execution (all traces).

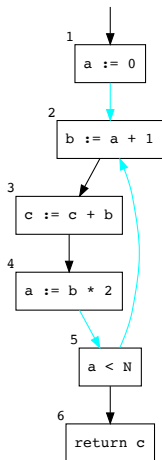
Remark: $\text{pred}[n]$ (resp. $\text{succ}[n]$) denotes the predecessors (resp. successors) of node n

Flow Graph for the example

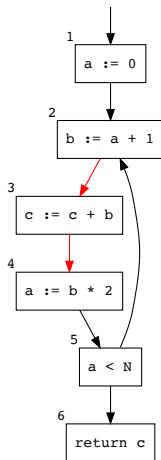
```
1      a := 0
2  L1:  b := a + 1
3      c := c + b
4      a := b * 2
5      if a < N goto L1
6      return c
```



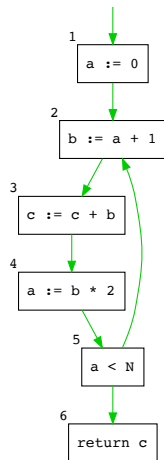
Liveness



Liveness for a



Liveness for b



Liveness for c

Terminology

- **defs:** nodes that define a variable, i.e. left (lhs) part of assignment nodes.
- **uses:** nodes that use (read) a variable (rhs).
- **live-in:** a variable is live-in of a node n if it lives on any in-edge of that node.
- **live-out:** a variable is live-out of a node n if it lives on any out-edge of that node.

Liveness computation

- 1 If a variable is in $use[n]$ (used by the node n), then it is live-in at node n .
- 2 If a variable is live-in at node n then it is live-out at all nodes in $pred[n]$.
- 3 If a variable is live-out at node n and not in $def[n]$ (not defined in node n), then it is live-in at node n (it must come from before).

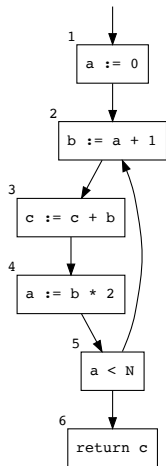
Dataflow Equations for Liveness Analysis

$$\begin{aligned}\text{in}[n] &= \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n]) \\ \text{out}[n] &= \bigcup_{s \in \text{succ}[n]} \text{in}[s]\end{aligned}$$

Possible Implementation (quadratic)

```
foreach  $n$   
   $\text{in}[n] \leftarrow \{ \}$   
   $\text{out}[n] \leftarrow \{ \}$   
  
repeat  
  foreach  $n$   
     $\text{in\_t}[n] \leftarrow \text{in}[n]$   
     $\text{out\_t}[n] \leftarrow \text{out}[n]$   
     $\text{in}[n] \leftarrow \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$   
     $\text{out}[n] \leftarrow \bigcup_{s \in \text{succ}[n]} \text{in}[s]$   
until  $\text{in\_t}[n] = \text{in}[n]$  and  $\text{out\_t}[n] = \text{out}[n]$  ( $\forall n$ )
```

Liveness Calculation : Iteration 1

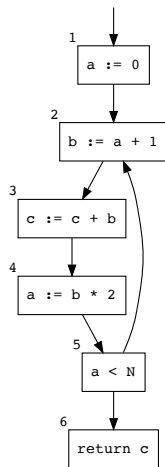


$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		
2	a	b	3		
3	bc	c	4		
4	b	a	5		
5	a		2,6		
6	c				

Liveness Calculation : Iteration 1

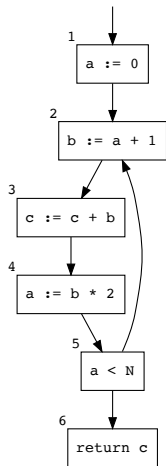


$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		
2	a	b	3	a	
3	bc	c	4		
4	b	a	5		
5	a		2,6		
6	c				

Liveness Calculation : Iteration 1

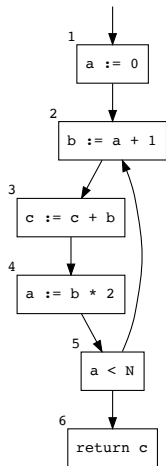


$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		
2	a	b	3	a	
3	bc	c	4	bc	
4	b	a	5		
5	a		2,6		
6	c				

Liveness Calculation : Iteration 1

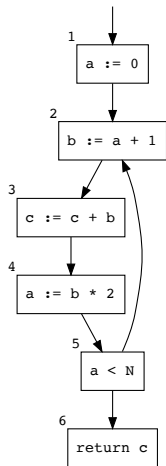


$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		
2	a	b	3	a	
3	bc	c	4	bc	
4	b	a	5	b	
5	a		2,6		
6	c				

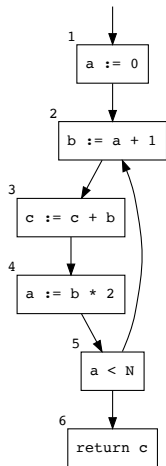
Liveness Calculation : Iteration 1



$$\begin{aligned} \text{in}[n] &= \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n]) \\ \text{out}[n] &= \bigcup_{s \in \text{succ}[n]} \text{in}[s] \end{aligned}$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		
2	a	b	3	a	
3	bc	c	4	bc	
4	b	a	5	b	
5	a		2,6	a	
6	c				

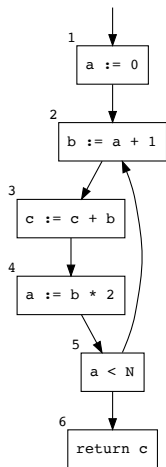
Liveness Calculation : Iteration 1



$$\begin{aligned} \text{in}[n] &= \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n]) \\ \text{out}[n] &= \bigcup_{s \in \text{succ}[n]} \text{in}[s] \end{aligned}$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		
2	a	b	3	a	
3	bc	c	4	bc	
4	b	a	5	b	
5	a		2,6	a	a
6	c				

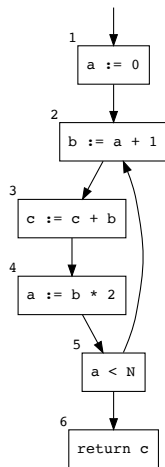
Liveness Calculation : Iteration 1



$$\begin{aligned} \text{in}[n] &= \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n]) \\ \text{out}[n] &= \bigcup_{s \in \text{succ}[n]} \text{in}[s] \end{aligned}$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		
2	a	b	3	a	
3	bc	c	4	bc	
4	b	a	5	b	
5	a		2,6	a	a
6	c			c	

Liveness Calculation : Iteration 2

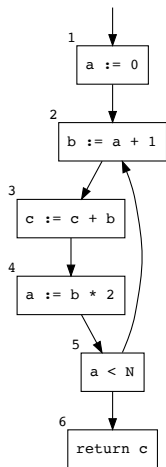


$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		
2	a	b	3	a	
3	bc	c	4	bc	
4	b	a	5	b	
5	a		2,6	a	a
6	c			c	

Liveness Calculation : Iteration 2

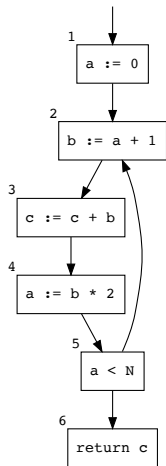


$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		a
2	a	b	3	a	
3	bc	c	4	bc	
4	b	a	5	b	
5	a		2,6	a	a
6	c			c	

Liveness Calculation : Iteration 2

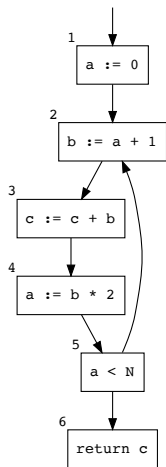


$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		a
2	a	b	3	a	bc
3	bc	c	4	bc	
4	b	a	5	b	
5	a		2,6	a	a
6	c			c	

Liveness Calculation : Iteration 2

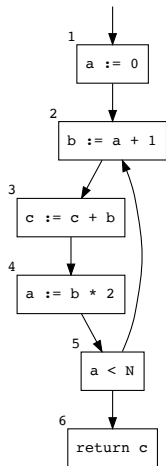


$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		a
2	a	b	3	a	bc
3	bc	c	4	bc	b
4	b	a	5	b	
5	a		2,6	a	a
6	c			c	

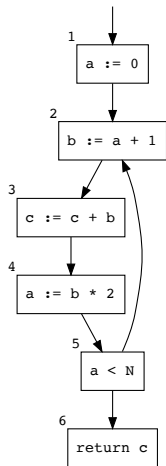
Liveness Calculation : Iteration 2



$$\begin{aligned} \text{in}[n] &= \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n]) \\ \text{out}[n] &= \bigcup_{s \in \text{succ}[n]} \text{in}[s] \end{aligned}$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		a
2	a	b	3	a	bc
3	bc	c	4	bc	b
4	b	a	5	b	a
5	a		2,6	a	a
6	c			c	

Liveness Calculation : Iteration 2



$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

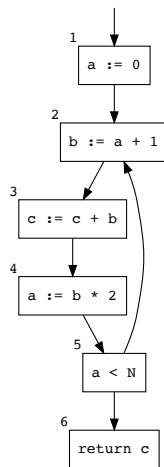
$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>
1		a	2		a
2	a	b	3	a	bc
3	bc	c	4	bc	b
4	b	a	5	b	a
5	a		2,6	a	ac
6	c			c	

Liveness Calculation Fixpoint

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2								
2	a	b	3								
3	bc	c	4								
4	b	a	5								
5	a		2,6								
6	c										

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2						
2	a	b	3						
3	bc	c	4						
4	b	a	5						
5	a		2,6						
6	c								



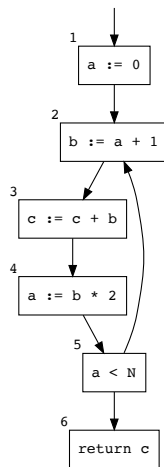
$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

Liveness Calculation Fixpoint

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	1st step							
				<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2								
2	a	b	3	a							
3	bc	c	4	bc							
4	b	a	5	b							
5	a		2,6	a	a						
6	c			c							

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2						
2	a	b	3						
3	bc	c	4						
4	b	a	5						
5	a		2,6						
6	c								



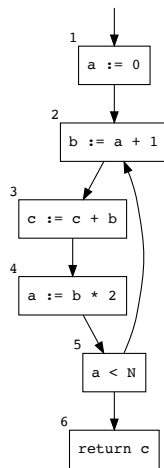
$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

Liveness Calculation Fixpoint

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	1st step		2nd step					
				<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2				a				
2	a	b	3	a		a	bc				
3	bc	c	4	bc		bc	b				
4	b	a	5	b		b	a				
5	a		2,6	a	a	a	ac				
6	c			c		c					

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2						
2	a	b	3						
3	bc	c	4						
4	b	a	5						
5	a		2,6						
6	c								



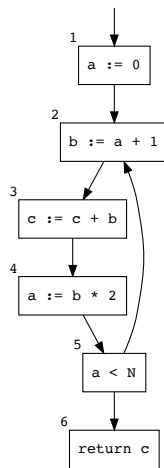
$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

Liveness Calculation Fixpoint

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	1st step		2nd step		3rd step			
				<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2				a		a		
2	a	b	3	a		a	bc	ac	bc		
3	bc	c	4	bc		bc	b	bc	b		
4	b	a	5	b		b	a	b	a		
5	a		2,6	a	a	a	ac	ac	ac		
6	c			c		c		c			

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2						
2	a	b	3						
3	bc	c	4						
4	b	a	5						
5	a		2,6						
6	c								



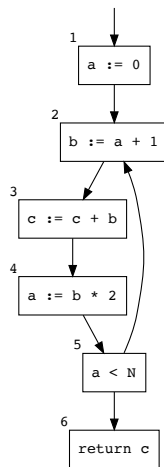
$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

Liveness Calculation Fixpoint

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	1st step		2nd step		3rd step		4th step	
				<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2				a		a	ac	
2	a	b	3	a		a	bc	ac	bc	ac	bc
3	bc	c	4	bc		bc	b	bc	b	bc	b
4	b	a	5	b		b	a	b	a	b	ac
5	a		2,6	a	a	a	ac	ac	ac	ac	ac
6	c			c		c		c		c	

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2						
2	a	b	3						
3	bc	c	4						
4	b	a	5						
5	a		2,6						
6	c								



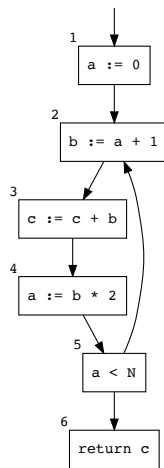
$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

Liveness Calculation Fixpoint

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	1st step		2nd step		3rd step		4th step	
				<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2				a		a	ac	
2	a	b	3	a		a	bc	ac	bc	ac	bc
3	bc	c	4	bc		bc	b	bc	b	bc	b
4	b	a	5	b		b	a	b	a	b	ac
5	a		2,6	a	a	a	ac	ac	ac	ac	ac
6	c			c		c		c		c	

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	5th step		<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
				<i>in</i>	<i>out</i>				
1		a	2	c	ac				
2	a	b	3	ac	bc				
3	bc	c	4	bc	b				
4	b	a	5	bc	ac				
5	a		2,6	ac	ac				
6	c			c					



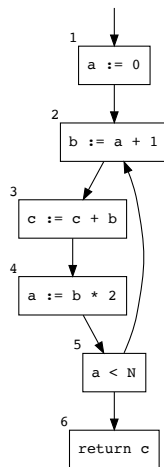
$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

Liveness Calculation Fixpoint

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	1st step		2nd step		3rd step		4th step	
				<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2				a		a	ac	
2	a	b	3	a		a	bc	ac	bc	ac	bc
3	bc	c	4	bc		bc	b	bc	b	bc	b
4	b	a	5	b		b	a	b	a	b	ac
5	a		2,6	a	a	a	ac	ac	ac	ac	ac
6	c			c		c		c		c	

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	5th step		6th step			
				<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2	c	ac	c	ac		
2	a	b	3	ac	bc	ac	bc		
3	bc	c	4	bc	b	bc	bc		
4	b	a	5	bc	ac	bc	ac		
5	a		2,6	ac	ac	ac	ac		
6	c			c		c			



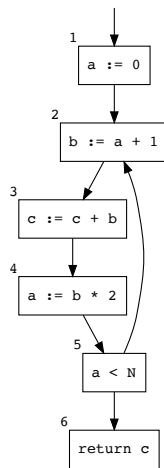
$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

Liveness Calculation Fixpoint

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	1st step		2nd step		3rd step		4th step	
				<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2				a		a	ac	
2	a	b	3	a		a	bc	ac	bc	ac	bc
3	bc	c	4	bc		bc	b	bc	b	bc	b
4	b	a	5	b		b	a	b	a	b	ac
5	a		2,6	a	a	a	ac	ac	ac	ac	ac
6	c			c		c		c		c	

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	5th step		6th step		7th step	
				<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2	c	ac	c	ac	c	ac
2	a	b	3	ac	bc	ac	bc	ac	bc
3	bc	c	4	bc	b	bc	bc	bc	bc
4	b	a	5	bc	ac	bc	ac	bc	ac
5	a		2,6	ac	ac	ac	ac	ac	ac
6	c			c		c		c	

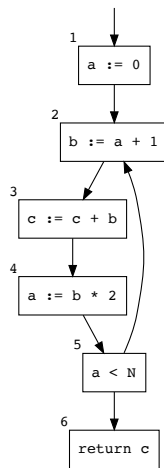


$$\begin{aligned} \text{in}[n] &= \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n]) \\ \text{out}[n] &= \bigcup_{s \in \text{succ}[n]} \text{in}[s] \end{aligned}$$

Liveness Calculation Fixpoint(Forward)

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	1st step		2nd step		3rd step		4th step	
				<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2				a		a	ac	
2	a	b	3	a		a	bc	ac	bc	ac	bc
3	bc	c	4	bc		bc	b	bc	b	bc	b
4	b	a	5	b		b	a	b	a	b	ac
5	a		2,6	a	a	a	ac	ac	ac	ac	ac
6	c			c		c		c		c	

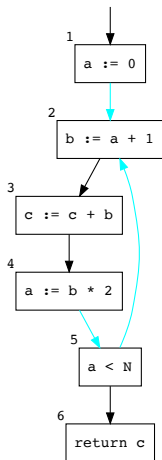
<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	5th step		6th step		7th step	
				<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>
1		a	2	c	ac	c	ac	c	ac
2	a	b	3	ac	bc	ac	bc	ac	bc
3	bc	c	4	bc	b	bc	bc	bc	bc
4	b	a	5	bc	ac	bc	ac	bc	ac
5	a		2,6	ac	ac	ac	ac	ac	ac
6	c			c		c		c	



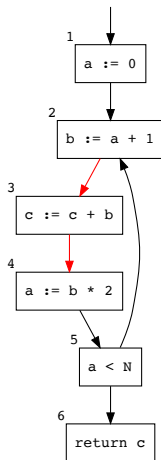
$$\text{in}[n] = \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n])$$

$$\text{out}[n] = \bigcup_{s \in \text{succ}[n]} \text{in}[s]$$

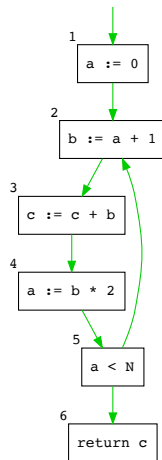
Liveness



Liveness for a



Liveness for b



Liveness for c

Why Liveness Analysis is Backward (Not Forward)

- **Liveness:** A variable is *live* at a point if it may be used **later** before being redefined.

Why Liveness Analysis is Backward (Not Forward)

- **Liveness:** A variable is *live* at a point if it may be used **later** before being redefined.
- **Forward is awkward:**
 - ▶ Must predict **future usage** from the current point
 - ▶ Requires tracking all *future paths* and possible needs
 - ▶ More state, more merging — inefficient and unintuitive

Why Liveness Analysis is Backward (Not Forward)

- **Liveness:** A variable is *live* at a point if it may be used **later** before being redefined.
- **Forward is awkward:**
 - ▶ Must predict **future usage** from the current point
 - ▶ Requires tracking all *future paths* and possible needs
 - ▶ More state, more merging — inefficient and unintuitive
- **Backward is natural:**
 - ▶ Looks **from uses back to definitions**
 - ▶ At each point: “What will be needed *after* this?”
 - ▶ Simple, local dataflow: propagate use and def sets backward

Liveness Calculation (Backward)

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>
6	c								
5	a		2,6						
4	b	a	5						
3	bc	c	4						
2	a	b	3						
1		a	2						

$$\begin{aligned} \text{in}[n] &= \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n]) \\ \text{out}[n] &= \bigcup_{s \in \text{succ}[n]} \text{in}[s] \end{aligned}$$

Calculation done following *reverse* control-flow edges.

Liveness Calculation (Backward)

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	1st step					
				<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>
6	c				c				
5	a		2,6	c	ac				
4	b	a	5	ac	bc				
3	bc	c	4	bc	bc				
2	a	b	3	bc	ac				
1		a	2	ac	c				

$$\begin{aligned} \text{in}[n] &= \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n]) \\ \text{out}[n] &= \bigcup_{s \in \text{succ}[n]} \text{in}[s] \end{aligned}$$

Calculation done following *reverse* control-flow edges.

Liveness Calculation (Backward)

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	1st step		2nd step		<i>out</i>	<i>in</i>
				<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>		
6	c				c		c		
5	a		2,6	c	ac	ac	ac		
4	b	a	5	ac	bc	ac	bc		
3	bc	c	4	bc	bc	bc	bc		
2	a	b	3	bc	ac	bc	ac		
1		a	2	ac	c	ac	c		

$$\begin{aligned} \text{in}[n] &= \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n]) \\ \text{out}[n] &= \bigcup_{s \in \text{succ}[n]} \text{in}[s] \end{aligned}$$

Calculation done following *reverse* control-flow edges.

Liveness Calculation (Backward)

<i>n</i>	<i>use</i>	<i>def</i>	<i>succ</i>	1st step		2nd step		3rd step	
				<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>
6	c				c		c		c
5	a		2,6	c	ac	ac	ac	ac	ac
4	b	a	5	ac	bc	ac	bc	ac	bc
3	bc	c	4	bc	bc	bc	bc	bc	bc
2	a	b	3	bc	ac	bc	ac	bc	ac
1		a	2	ac	c	ac	c	ac	c

$$\begin{aligned} \text{in}[n] &= \text{use}[n] \cup (\text{out}[n] \setminus \text{def}[n]) \\ \text{out}[n] &= \bigcup_{s \in \text{succ}[n]} \text{in}[s] \end{aligned}$$

Calculation done following *reverse* control-flow edges.

Summary

